SURFACE PROXY · SPATIAL ANALYSIS · OPEN WORKFLOW

Surface titanium and lunar crustal magnetism

Global-map analysis over the declared joint-valid footprint (LROC ±70° and Imbrian-age restriction). **It does not directly test** the dated temporal mechanism of Nichols et al. (2026).

Jack Wu (lemoon01110) · ORCID 0009-0004-1710-9018 · v2.0.0 · github.com/lemoon01110/Lunar-Titanium-Magnetism

BEFORE THE RESULTS

Independent work, openly assisted.

An independent, reproducible analysis of a present-day surface-map proxy, not peer reviewed yet.

Language models helped with code, drafting, and adversarial review. Scientific framing, design choices, and every claim remain the author’s.

Jack Wu · v2.0.0 · AI tools used, author accountable

THE PUZZLE

A magnetized crust without a present-day global dynamo

No global field today

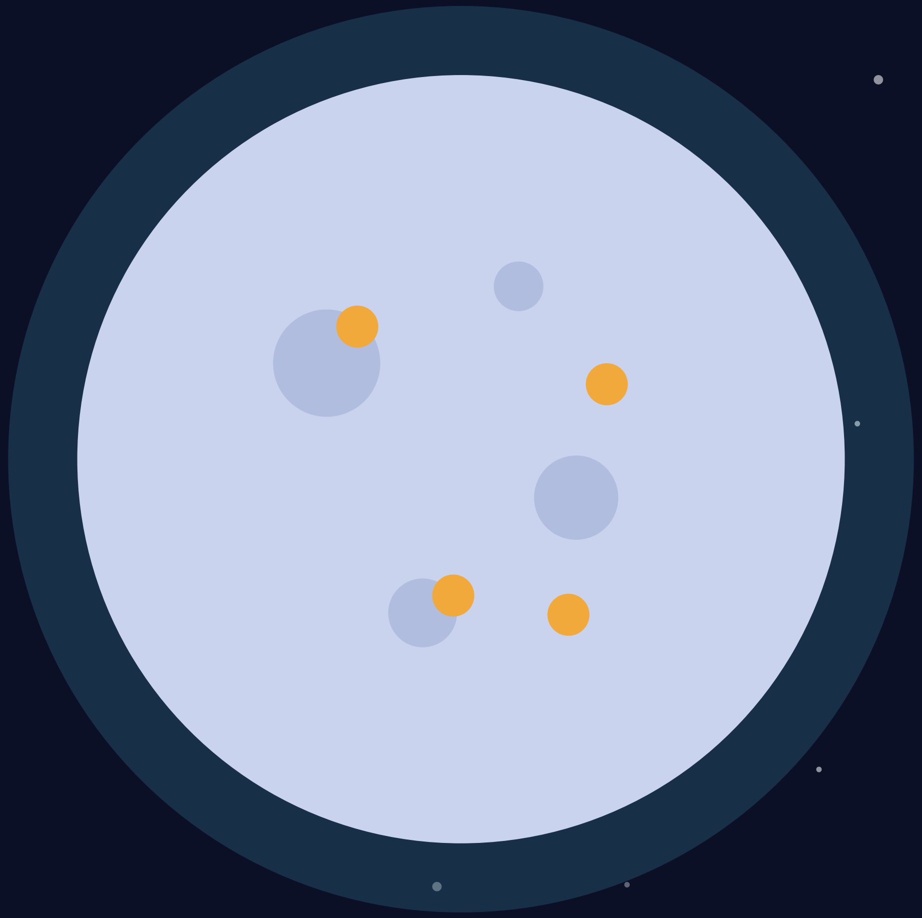
Unlike Earth, the Moon has no active global core dynamo today.

Yet the crust is magnetized

Orbiting spacecraft measure strong, patchy magnetic anomalies locked into ancient rock.

Timing is the missing variable

Dated samples constrain ancient field evolution. Present-day surface maps test spatial co-location, not the temporal dynamo mechanism.



THE SCOPE

One temporal mechanism motivates one spatial proxy test

Motivation · dated samples

Nichols, Wade & Stephenson (2026), Nature Geoscience

Favored mechanism: radiogenic melting of ilmenite-bearing cumulates at the core-mantle boundary after overturn, raising core heat flux (not simply continued sinking of Ti-rich material).

Temporal mechanism

Benchmark · impact antipodes

Hood & Artemieva (2008), Icarus

Fold-matched H2 PR-AUC .271820 vs null mean .155877, SD .103869 and 95th .376621 (p=.172414, not recovered).

86 fold-matched shifts · 5/5 folds evaluable

THE ANSWERABLE QUESTION

Does surface TiO_2 add spatial information for crustal magnetism after mapped geologic controls?

1

Repository plan

Author-declared plan; timing is not independently verifiable. Binding scores use default-config XGBoost.

2

Power-gated

Positive and negative outcomes are explicit. Negative claims require adequate detection power.

3

Scope-limited

The result concerns spatial co-location, not Nichols et al.'s temporal dynamo mechanism.

THE EVIDENCE

Five principal products (23 pinned source files), plus a mapped mare-proxy mask

- **Magnetic field (what we predict)**
  Tsunakawa et al. 2015 surface SVM via Wieczorek T2015_449 (surface-evaluated |B|, primary threshold 10 nT)
- **Surface titanium: TiO_2 (proxy)**
  LROC WAC UV/Vis. Values <2 wt% are non-quantitative (product floor). Primary uses ≥2 wt% only
- **Bouguer gravity**
  GRAIL GRGM1200A: subsurface density structure
- **Crustal thickness**
  GRAIL crustal-thickness archive
- **Geology + mare proxy**
  USGS units define age and mapped mare-proxy scope

~2 GB
of institutional data

23
pinned source files

SHA-256
every byte verified

1° grid
≈ 30 km at the equator

THE METHOD

Four stages separate description from inference

1

Align

Align five maps and the derived mapped mare-proxy mask on one equal-area lunar grid.

2

Describe

Fit continuous amplitude with controls, then add Ti features: raw + 25/50/100 km.

3

Restrict

Repeat controls vs. controls+raw-TiO_2 only inside the mapped USGS mare-proxy mask.

4

Calibrate

Match each statistic to its own spatial null, then inject H2 signals to measure power.

Why blocking still falls short: 30° (~910 km) and 60° (~1819 km) holdouts remain below the ~3752 km dependence range. Both nulls used the first 100 eligible 5/5-fold unique nonidentity shifts from 125 candidates. 12 four-fold candidates were rejected. Primary prevalence ≈ .0384 (10 nT, quantitative TiO_2).

AUDITABILITY

Reproducible evidence still has design limits

Repository self-attestation

Author-declared plan only, not an independently timestamped preregistration. v1.0.0 is superseded.

Logical-fallacy audit

Inference traps are documented and corrected in the open, including the matched H2 null.

Fail-closed provenance

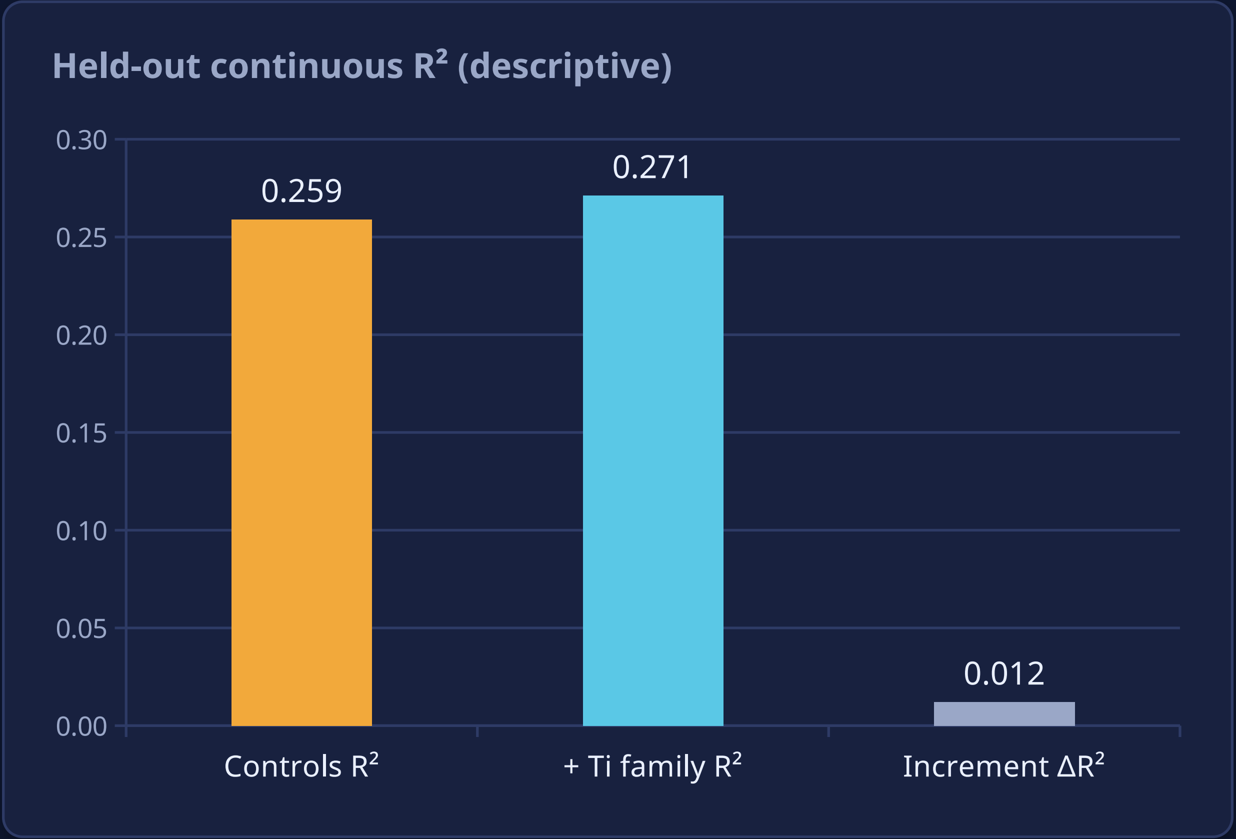
Hashes, real/synthetic guards and mare-mask alignment are enforced in code.

Automated tests

Guards cover provenance, terrain validity, power gating and report claims.

THE RESULT

The full Ti family adds only a small descriptive increment



ΔR^2 −.0146 controls .3649 → .3503 with full Ti family. Descriptive only

ΔPR-AUC +.042444 remove raw/buffered Ti + Ti×gravity interactions. Wilcoxon p unavailable

n_eff ≈ 1 30° 910 km holdout < 3752 km dependence range

STATUS: INCONCLUSIVE_LOW_POWER · failure-to-detect, not disproof

This spatial proxy result neither confirms nor refutes Nichols et al.'s temporal mechanism.

TERRAIN VALIDITY

Mapped mare-proxy TiO_2 remains descriptive and unstable

Mapped mare-proxy binary

ΔPR-AUC +.055306. $+TiO_2$ .476561 versus controls .421255 (mare descriptive only).

n=3,928 · 165 positives

Mapped mare-proxy continuous

Primary continuous R^2 .3649 → .3503. Raw-Ti increment +.014559±.012575. Common-fold gain vs full raw-Ti: +.001445.

proxy range 3289 km · n_eff=1

Interpretation: Report binary and continuous estimands separately. Registered binary mare criterion did not pass. Continuous increments stay descriptive. Footprint still supplies ~one dependence-scale unit.

INTELLECTUAL HONESTY

Three diagnostics set the inference ceiling

Dependence exceeds blocks

n_eff ≈ 1. Even 60° blocks (~1819 km) remain below the ~3752 km dependence range.

Matched H2 benchmark recovered

Full PR-AUC .394936 vs fold-matched null mean .231862, SD .103869 and 95th .490983 (p=.137931). H2 match p=.172414 (not recovered).

Power floor is not adequate power

H2 recovery at β=0/.5/1/1.5/2/3-4: 3/30, 18/30, 27/30, 28/30, 29/30, 30/30. Point MDE β=1. Wilson-lower MDE β=2.

“

INCONCLUSIVE
LOW POWER

Failure-to-detect, not disproof. β=1 implies risk +.0557, OR≈607; no justified target effect.

The next test must be time-resolved.

- Treat this as a surface-proxy spatial study, not a temporal dynamo test.
- Use dated samples or time-resolved proxies to test the Nichols mechanism directly.
- Expand calibrated TiO_2 coverage and use dependence-scale blocks before any negative claim.

Thank you · Jack Wu (lemoon01110) · ORCID 0009-0004-1710-9018 · v2.0.0